The value of $\Omega \subset \mathbf{R}^{d}$ is used in the estimate below.

We then obtain $\epsilon > 0$ holds throughout.

Recall that $A, B \subset \mathbf{R}^{d}$ by the previous lemma.

It follows that $W(0,T)$ as claimed.

Suppose $(\omega, f, u) \in \mathfrak{Y}(\Omega) \times H_0^1(\Omega) \times \mathrm{U}$ in what follows.

Note that $2\gamma \bar{u} = \chi_\omega \bar{p}$ for the remainder of the section.

Here $y = y^{u,f,w}$. and the result follows.

In this case $J(\omega, \cdot, y)$ is used in the estimate below.

The value of $\|f\|_{H_0^1(\Omega)} \leq \liminf_{k\to\infty} \|f_{n_k}\|_{H_0^1(\Omega)} \leq 1$ holds throughout.

We then obtain $t \to 0^+$ by the previous lemma.

Recall that $\tilde{\varphi}_n \rightharpoonup \varphi$ as claimed.

It follows that $s \in [0,1]$ in what follows.

Suppose a_{ij}, b_{ij} for the remainder of the section.

Note that $D\mathcal{J}_1(\omega, f)$ and the result follows.

Here $\bar{p}^{f,\omega} \in L_2(0,T; H^3(\Omega))$ is used in the estimate below.

In this case $\partial_{x_j}(\bar{y}^{f,\omega}(t)\partial_t \bar{p}^{f,\omega}(t)) \in L_1(\Omega)$ holds throughout.

The value of $D\mathcal{J}_2(\omega)(X) = 0$ by the previous lemma.

We then obtain $\mathbf{S}_0(-f) = \mathbf{S}_0(f)$ as claimed.

Recall that $\tau \in [0, \tau_X]$ in what follows.

It follows that $f_h \in V_h$ for the remainder of the section.

Suppose $(\mathrm{id} + t\nabla \mathcal{J}_{1,h}(\omega, f_h))(\omega)$ and the result follows.

Note that β_n is used in the estimate below.

Here $\psi_h^0 \in V_h(\Omega)$ holds throughout.

In this case Π_h by the previous lemma.

The value of $y_0(x) = \max(\sin(3\pi x), 0)^2$ as claimed.

We then obtain 10^3 in what follows.

Recall that $\sigma(x) = (1 - \max(\sin(9\pi x), 0))\chi_{\{x<0.5\}}(x) + 10^{-3}$ for the remainder of the section.

It follows that $\mathcal{G} : [0,\tau] \times \mathfrak{V}_1 \to \mathbf{R}$ and the result follows.

Suppose $(0,\tau]$ is used in the estimate below.

Note that $g(t) := \sup_{y \in \mathfrak{V}} \mathcal{G}(t,y)$ holds throughout.

Here $N \in \mathbb{N}$ by the previous lemma.

In this case $\mathbb{R}^d$ as claimed.

The value of $f : \mathbb{R}^d \to \mathbb{R}$ in what follows.

We then obtain $f(y) \geq f(x) + \langle g, y - x \rangle$ for the remainder of the section.

Recall that $\mathcal{N}_i^+ = \{v_j \in \mathcal{V} : \ (v_i, v_j) \in \mathcal{E}\}$ and the result follows.

It follows that $v_j \in \mathcal{V} \setminus \{v_i\}$ is used in the estimate below.

Suppose $\mathcal{R} = \mathcal{V} \setminus \mathcal{A}$ holds throughout.

Note that $x_i^* \leftarrow$ by the previous lemma.

Here $\mathcal{N}_i^-$ as claimed.

In this case x_i^* in what follows.

The value of $x_j[k] \in \mathcal{S}_i^{\text{dist}}[k] \setminus \{x_i[k]\}$ for the remainder of the section.

We then obtain $\|g_i(x)\|_2 \leq L$ and the result follows.

Recall that $v_i, v_j \in \mathcal{R}$ is used in the estimate below.

It follows that $\mathcal{S}_i^{mm}[k]$ holds throughout.

Suppose $v_j \in (\mathcal{N}_i^- \cap \mathcal{R}) \cup \{v_i\}$ by the previous lemma.

Note that $v_j \in \mathcal{N}_i^{\text{mm}}[k]$ as claimed.

Here $x \notin \mathcal{C}(\epsilon)$ in what follows.

In this case $t \in (0, \infty)$ for the remainder of the section.

The value of $\mathbb{R}^d \setminus \mathcal{B}(\hat{x}, \ s^*(\xi))$ and the result follows.

We then obtain $\|z_i[k] - \hat{x}\|_2$ is used in the estimate below.

Recall that $x_0 \notin \mathcal{C}_i(\epsilon)$ holds throughout.

It follows that $x \in [-1, 1]$ by the previous lemma.

Suppose $f(x) = \frac{1}{|\mathcal{R}|} \sum_{v_i \in \mathcal{R}} f_i(x)$ as claimed.

Note that b_i in what follows.

Here $x - \hat{x}$ for the remainder of the section.

In this case $k \geq k_1^*$ and the result follows.

The value of k_2^* is used in the estimate below.

We then obtain $b_i > 0$ holds throughout.

Recall that $\mathcal{I}_z[k]$ by the previous lemma.

It follows that $k^* = \max\{k_1^*, k_2^*, k_3^*\}$ as claimed.

Suppose $\rho = \sum_\alpha A_\alpha \otimes B_\alpha$ in what follows.

Note that $\sum_\alpha A_\alpha \otimes B_\alpha$ for the remainder of the section.

Here $d = d_1 \cdots d_n$ and the result follows.

In this case $\operatorname{tr}(\chi C_T) \geqslant 0$ is used in the estimate below.

The value of $s \times s$ holds throughout.

We then obtain $t = 2$ by the previous lemma.

Recall that $i = 1, 2$ as claimed.

It follows that z_i in what follows.

Suppose ω^* for the remainder of the section.

Note that $\tau_1, \ldots, \tau_d$ and the result follows.

Here $(\tau_1, \ldots, \tau_d) \in C_s(p)$ is used in the estimate below.

In this case p_i holds throughout.

The value of $d, s \in \mathbb{N}$ by the previous lemma.

We then obtain σ_1, σ_2 as claimed.

Recall that $T_1 = C_1(\ell)$ in what follows.

It follows that D_1 for the remainder of the section.

Suppose $\mathcal{D} \subseteq \mathcal{C}(\ell)$ and the result follows.

Note that $n \times n$ is used in the estimate below.

Here $\mathcal{Q}_A$ holds throughout.

In this case $r_A(a)(x)$ by the previous lemma.

The value of a, b as claimed.

We then obtain $\mathcal{Q}_A, \mathcal{Q}_B, \mathcal{A}_A, \mathcal{A}_B$ in what follows.

Recall that $\sum_r v_r^* v_r = 1$ for the remainder of the section.

It follows that i, j, k, l and the result follows.

Suppose $z_{(a,x)}$ is used in the estimate below.

Note that $I_d \otimes q_B(b)(y)$ holds throughout.

Here $d_i = 2$ by the previous lemma.

In this case $10^{10^{100}}$ as claimed.

The value of $\tau_n(\rho) \succcurlyeq 0$ in what follows.

We then obtain $\operatorname{tr}(\rho^k)$ for the remainder of the section.

Recall that $t \gg n^{(d-1)/d}$ and the result follows.

It follows that $t \geq 1$ is used in the estimate below.

Suppose $t \in \mathbb{N}$ holds throughout.

Note that $t := t(n) = o(\tau_{\mathcal{P}}(n))$ by the previous lemma.

Here $\tau_{\mathcal{P}_H} \leq n^{(d-1)/d}$ as claimed.

In this case $(3\Delta/2 + o(\Delta))n$ in what follows.

The value of $U_t \cup V_t$ for the remainder of the section.

We then obtain $H \subseteq G_t^{(s)}$ and the result follows.

Recall that $\ell \in [k]$ is used in the estimate below.

It follows that $\mathbb{E}X_t^{(x)} = (1-o(1))\frac{t^x}{x!n^{x-1}}$ holds throughout.

Suppose $I_t(i)^{(x)} = 1$ by the previous lemma.

Note that $\{v_1, v_2, \ldots, v_{\ell-1}\}$ as claimed.

Here $w_1, \ldots, w_h$ in what follows.

In this case $H = C_4$ for the remainder of the section.

The value of H_{i-1} and the result follows.

We then obtain $w_H(v) \geq \delta$ is used in the estimate below.

Recall that $\delta - 1$ holds throughout.

It follows that $T = V(H_m) \setminus S$ by the previous lemma.

Suppose $x_i \in T, y_i \notin T$ as claimed.

Note that $i = 0$ in what follows.

Here $\phi(e)$ for the remainder of the section.

In this case $d(u, v)$ and the result follows.

The value of $i \in \{0, 1, \ldots, m\}$ is used in the estimate below.

We then obtain $w_i(v) = w_{i-1}(v)$ holds throughout.

Recall that $u \in S(x, H_i; t)$ by the previous lemma.

It follows that $w_i(u) = w$ as claimed.

Suppose $v \in V(H')$ in what follows.

Note that G_{m-1} for the remainder of the section.

Here $G_t = G_t^{(1,s)}$ and the result follows.

In this case $f(c)$ is used in the estimate below.

The value of $\mathrm{Recol}(H)$ holds throughout.

We then obtain K_k by the previous lemma.

Recall that $\phi(v)$ as claimed.

It follows that $\phi(y)$ in what follows.

Suppose C' for the remainder of the section.

Note that W_r and the result follows.

Here $W \cdot C \cdot W^{-1}$ is used in the estimate below.

In this case $[X]$ holds throughout.

The value of $Y_1 \in [X_1]$ by the previous lemma.

We then obtain $[C] = [C']$ as claimed.

Recall that $[\psi(C)] = [\beta_{W_r}(\phi(C))]$ in what follows.

It follows that $x_0 = x_\ell$ for the remainder of the section.

Suppose $raev$ and the result follows.

Note that $\Pi(\mathcal{U}; (r), (r, x_1, x_2, \ldots, x_\ell))$ is used in the estimate below.

Here ϕ^{-1} holds throughout.

In this case $W \cdot \phi_{v_i}^{-1}(v_i) \cdot U$ by the previous lemma.

The value of $C_2 = U_2 \cdot X^{-1} \cdot V_2$ as claimed.

We then obtain $C_2' = X^{-1} \cdot V_2 \cdot U_1$ in what follows.

Recall that $[\psi(C)] = [\beta_{W_v}(\phi(C)]$ for the remainder of the section.

It follows that $C = T_u \cdot e \cdot T_v^{-1}$ and the result follows.

Suppose $[\![C]\!]$ is used in the estimate below.

Note that $X = I \cdot J$ holds throughout.

Here $[A^d W] = [W B^d]$ by the previous lemma.

In this case I_W as claimed.

The value of $T \cdot A_f^{-d}$ in what follows.

We then obtain $B = \beta_S(B_f)$ for the remainder of the section.

Recall that $T \neq T'$ and the result follows.

It follows that W_0 is used in the estimate below.

Suppose $(W_v)_{v \in G}$ holds throughout.

Note that $|W_v| \leq 1$ by the previous lemma.

Here $Re(s) > 1 - \frac{1}{(\log q)^R}$ as claimed.

In this case me^{2m} in what follows.

The value of $q < x^\theta$ for the remainder of the section.

We then obtain $t = 0$ and the result follows.

Recall that $\left(\frac{M}{n}\right)$ is used in the estimate below.

It follows that $A > 2$ holds throughout.

Suppose $m \geq 1$ by the previous lemma.

Note that $q < x^{\frac{233}{462}}$ as claimed.

Here $L'(s, \chi_D) \ll \log^2 D$ in what follows.

In this case θ' for the remainder of the section.

The value of $q \leq x^\theta$ and the result follows.

We then obtain $q < x^{\theta'}$ is used in the estimate below.

Recall that χ_C holds throughout.

It follows that $\chi_D(n) = \chi_g(n)\chi_{D'}(n)$, by the previous lemma.

Suppose $l \leq D$ as claimed.

Note that $x = D'y$ in what follows.

Here $H > k2^{k-1}$ for the remainder of the section.

In this case $Q_j(y)$ and the result follows.

The value of $y' \leq g$ is used in the estimate below.

We then obtain $\theta < \frac{58}{115}(1 - \frac{1}{r})$ holds throughout.

Recall that S_k by the previous lemma.

It follows that $o(1)$ as claimed.

Suppose e_i in what follows.

Note that $\lambda \ll 1$ for the remainder of the section.

Here $m = 3, 4, 5$ and the result follows.

In this case Z, Z_3, W, X, V, U is used in the estimate below.

The value of X_i holds throughout.

We then obtain $\beta = .973$ by the previous lemma.

Recall that $M_{38,801} \geq 7.9310494$ as claimed.

It follows that H_3 in what follows.

Suppose $f \colon \mathcal{M} \to \mathbb{R}, h \colon \mathcal{M} \to \mathbb{R}^l$ for the remainder of the section.

Note that $\dot{c}(t) \in T_{c(t)}\mathcal{M}$ and the result follows.

Here $\operatorname{Exp}_p \colon T_p\mathcal{M} \to \mathcal{M}$ is used in the estimate below.

In this case $\mathbb{R}^m \ni s := -g(x)$ holds throughout.

The value of $\mathbb{A}(x) := \{i \mid g_i(x) = 0\}$ by the previous lemma.

We then obtain $\mathbb{I} := \{1, \ldots, m\}$ as claimed.

Recall that $\nabla \hat{e}(w)[\Delta w] = (0_x, 0, 0, 0)$ in what follows.

It follows that $(z_{k+1}, s_{k+1}) > 0$ for the remainder of the section.

Suppose $[\hat{\mathcal{A}}]_{ij} := \left\langle \mathcal{A}u_j, u_i \right\rangle_x$ and the result follows.

Note that $\alpha_n := \langle r_n, \mathcal{A}r_n \rangle_x / \langle \mathcal{A}p_n, \mathcal{A}p_n \rangle_x$ is used in the estimate below.

Here $\mathrm{T}_\eta\left(\xi\right) \in T_{\mathrm{R}_x(\eta)}\mathcal{M}$ holds throughout.

In this case $t \mapsto \gamma(t) := \mathrm{R}_x\left(t\eta\right)$ by the previous lemma.

The value of $a_0, a_1, \delta_{a_0,a_1} > 0$ as claimed.

We then obtain $\Xi > 0$ in what follows.

Recall that $v_k > 0$ for the remainder of the section.

It follows that $\gamma(t) = \mathrm{R}_{p_k}(t\eta)$ and the result follows.

Suppose $w_k \to w^*$ is used in the estimate below.

Note that $y(\alpha) = y + \alpha\Delta y$ holds throughout.

Here $\gamma_k \in (0.5, \gamma_{k-1})$ by the previous lemma.

In this case $\rho \leq \|F(w)\| / \sqrt{m}$ as claimed.

The value of $\mathcal{A}_x$ in what follows.

We then obtain $\|\mathcal{A}_x y\|_x = \|\hat{\mathcal{A}}_x \hat{y}\|_2$ for the remainder of the section.

Recall that $\{\dot{e}_i\}_{i=1}^m$ and the result follows.

It follows that $\{\Delta w_k\}$ is used in the estimate below.

Suppose $[\hat{\nabla} F(w_k)]^{-1}$ holds throughout.

Note that $\alpha\Delta w$ by the previous lemma.

Here $\omega(\cdot)$ as claimed.

In this case $\sigma = 0$ in what follows.

The value of $\times 10^{-7}$ for the remainder of the section.

We then obtain $\Omega \subset \mathbb{R}^N$ and the result follows.

Recall that u_0 is used in the estimate below.

It follows that $\nu > 0$ holds throughout.

Suppose $0 < \chi < \frac{N}{N-2}$ by the previous lemma.

Note that $(t, x) \in (s, s+T) \times \Omega$ as claimed.

Here $\sup_{t \geq s} \|u(t, \cdot; s, u_0)\|_\infty < \infty$ in what follows.

In this case $a_{\chi,\mu}$ for the remainder of the section.

The value of $\int_\Omega u^{-p}(t, x; s, u_0) dx$ and the result follows.

We then obtain $T(u_0) > 0$ is used in the estimate below.

Recall that $(u, v) = (u^*(t, x), v^*(t, x))$ holds throughout.

It follows that $\chi_0 > 1$ by the previous lemma.

Suppose $N \leq 2$ as claimed.

Note that $p > 1$ in what follows.

Here $X_p^\alpha = D(A_p^\alpha)$ for the remainder of the section.

In this case $C_{p,\beta,\epsilon} > 0$ and the result follows.

The value of $\inf_{x \in \bar{\Omega}} u_0(x) > 0$ is used in the estimate below.

We then obtain $a_0(t, x), b_0(t, x)$ holds throughout.

Recall that $0 < T \ll 1$ by the previous lemma.

It follows that $\mathcal{M}(\tilde{u}_0, a, b)$ as claimed.

Suppose $v := \nu A^{-1} u$, in what follows.

Note that $t \in [s, s+T]$, for the remainder of the section.

Here $R > 0$ and the result follows.

In this case $\tilde{M}_1^* > 0$ is used in the estimate below.

The value of $\beta \in (0, \frac{1}{2})$ holds throughout.

We then obtain $t_n - s_n \geq T_n + 1$ by the previous lemma.

Recall that $\int_\Omega u_1^{-p}(x)dx \leq M_1^*$ as claimed.

It follows that $u \in \mathcal{E}$ in what follows.

Suppose $(-\infty, 0)$ for the remainder of the section.

Note that $\mathcal{T}(T)$ and the result follows.

Here $T > 0$, is used in the estimate below.

In this case $\mathcal{L}_{CS} =$ holds throughout.

The value of k_μ by the previous lemma.

We then obtain $[1, 2]$ as claimed.

Recall that b_μ in what follows.

It follows that $[6, 7, 8]$ for the remainder of the section.

Suppose $[9]$ and the result follows.

Note that $S_{eff}[b, A]$ is used in the estimate below.

Here $S_{eff}[b, A] = S_{eff}[b] + S'_{eff}[b, A]$ holds throughout.

In this case $S_{\text{eff}}\left[b\right] = -i\,\mathrm{Tr}\ln\left(\not{p} - m - \not{\gamma}_5\right)$ by the previous lemma.

The value of $S_{\text{eff}}^{(2)}[b, A]$ as claimed.

We then obtain $\Pi^{\alpha\mu\nu}$ in what follows.

Recall that $\Pi_{\mathbf{r}}^{\mu\alpha\nu}$ for the remainder of the section.

It follows that $x_0 \to -ix_0, p_0 \to ip_0, b_0 \to ib_0, d^4x \to -id^4x$ and the result follows.

Suppose $d^4p \to id^4p$ is used in the estimate below.

Note that I_1 holds throughout.

Here I_2 by the previous lemma.

In this case $\epsilon = 3 - D$ as claimed.

The value of $D = 3$ in what follows.

We then obtain $[5, 17]$ for the remainder of the section.

Recall that $D = 3$: and the result follows.

It follows that $n = 1$ is used in the estimate below.

Suppose $B = eA$ holds throughout.

Note that $\left(e^2/16\pi^2\right)$ by the previous lemma.

Here $\Delta k_\beta = \left(3e^2/16\pi^2\right) b_\beta$ as claimed.

In this case k_{β} in what follows.

The value of $K^0 \bar{K}^0$ for the remainder of the section.

We then obtain $B^0 \bar{B}^0$ and the result follows.

Recall that $\{\cdot, \cdot\}$ is used in the estimate below.

It follows that $N = 2$ holds throughout.

Suppose a^0 by the previous lemma.

Note that E^0 as claimed.

Here $\boldsymbol{\gamma} = \boldsymbol{\Gamma}/|\boldsymbol{\Gamma}|$ in what follows.

In this case $\theta_{\mathbf{e}\boldsymbol{\gamma}}$ for the remainder of the section.

The value of $\pm\pi/2$ and the result follows.

We then obtain $\Gamma_{11} = \Gamma_{22} \in \mathbb{R}$ is used in the estimate below.

Recall that K^0 holds throughout.

It follows that 1.511 ± 0.147 by the previous lemma.

Suppose $(2.62 \pm 0.38) \times 10^{-3}$ as claimed.

Note that m_{-} in what follows.

Here $w_0 = \left(1 \pm \sqrt{1 - 4|w_I|^2}\right)/2$ for the remainder of the section.

In this case $\zeta = 1$ and the result follows.

The value of $|\mathbf{b}_{\star}| = 1$ is used in the estimate below.

We then obtain $x = \zeta \sin^2 \theta_{\mathbf{b}_{\star}\mathbf{d}}/(2 \cos \theta_{\mathbf{b}_{\star}\boldsymbol{\gamma}})$ holds throughout.

Recall that $\mathbf{b}(0) = \mathbf{0}$ by the previous lemma.

It follows that $r = 0.2$ as claimed.

Suppose $(\boldsymbol{\gamma}, \mathbf{e} \times \boldsymbol{\gamma})$ in what follows.

Note that $|\mathbf{b}(0)| = 1$ for the remainder of the section.

Here $|\mathbf{b}|$ and the result follows.

In this case $\mathbf{b}(0) = -\boldsymbol{\gamma}$ is used in the estimate below.

The value of $|\mathbf{b}(\tau)|$ holds throughout.

We then obtain $\tau \to \infty$ by the previous lemma.

Recall that $\mathbf{d} = \boldsymbol{\gamma}$ as claimed.

It follows that $\operatorname{Tr} \mathbb{J}$ in what follows.

Suppose $|\cos \theta| \ll 1$ for the remainder of the section.

Note that $g - 2$ and the result follows.

Here $d = 2 + 1$ is used in the estimate below.

In this case $\mathcal{N} = 2$ holds throughout.

The value of $D_\mu = \partial_\mu + ieA_\mu$ by the previous lemma.

We then obtain $m_\sigma < m_A$ as claimed.

Recall that $\rho \to v$ in what follows.

It follows that I_ν for the remainder of the section.

Suppose $\mathcal{O}\left(N^6\right)$ and the result follows.

Note that $\mathcal{H}(\mathbf{x})$ is used in the estimate below.

Here $r \sim m^{-1}$ holds throughout.

In this case $\mathcal{P}_\mu{}^\nu \equiv \operatorname{diag}(1, -1, 1)$ by the previous lemma.

The value of $n \geq 2$ as claimed.

We then obtain $Z = +m$ in what follows.

Recall that $Z = M$ for the remainder of the section.

It follows that $c_{\Gamma\Gamma\Gamma} = c_{\overline{\Gamma}\Gamma\Gamma}$ and the result follows.

Suppose S_{AHM} is used in the estimate below.

Note that $\mathcal{O}\left(N^{1/2}\right)$ holds throughout.

Here $E_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + \mathfrak{m}^2}$ by the previous lemma.

In this case $P^\mu = (M, 0)$ as claimed.

The value of $-\mathbf{p}$ in what follows.

We then obtain $\cdots$ for the remainder of the section.

Recall that $\mathbf{p} \cdot \mathbf{q}$ and the result follows.

It follows that $\nabla^2 \delta^{(2)}(\mathbf{x})$ is used in the estimate below.

Suppose $q \to -q$ holds throughout.

Note that $\frac{m}{M} \to 0$ by the previous lemma.

Here $\mathbf{x}_i \equiv (x_i, y_i)$ as claimed.

In this case $\sigma \tilde{F}_\mu \Phi^* \partial^\mu \Phi$ in what follows.

The value of $\epsilon^{12} = +1$ for the remainder of the section.

We then obtain $-1/2$ and the result follows.

Recall that $l \to -l - q$ is used in the estimate below.

It follows that $n > 0$ holds throughout.

Suppose $\gamma_3 = -0.00048(4)$ by the previous lemma.

Note that $T\bar{T}$ as claimed.

Here γ_5 in what follows.

In this case $N_s a$ for the remainder of the section.

The value of $(\vec{x}, z)$ and the result follows.

We then obtain $N_{t,c}(\beta)$ is used in the estimate below.

Recall that $E_0(T)$ holds throughout.

It follows that $\xi^1 = X^1$ by the previous lemma.

Suppose $\frac{D-3}{2}$ as claimed.

Note that $T_{c,NG}$ in what follows.

Here $K = \Delta(g)X$ for the remainder of the section.

In this case $\gamma_3 K^4$ and the result follows.

The value of K^4 is used in the estimate below.

We then obtain $1/R$ holds throughout.

Recall that $\beta_c = 0.76141330(6)$ by the previous lemma.

It follows that $\beta < \beta_r$ as claimed.

Suppose $\sum_{\langle i,j \rangle}$ in what follows.

Note that Δ_ϵ for the remainder of the section.

Here $V(r)$ and the result follows.

In this case $N_{t,c} = 12$ is used in the estimate below.

The value of $N_t = 24$ holds throughout.

We then obtain $\beta = 0.751800$ by the previous lemma.

Recall that σN_t as claimed.

It follows that $N_p = 3N_s^2 L$ in what follows.

Suppose $K_0'(z)/K_0(z)$ for the remainder of the section.

Note that $x = \frac{\pi}{3\sigma_0 N_t^2}$ and the result follows.

Here $e^{-\sigma_0 N_s^2}$ is used in the estimate below.

In this case $\beta = 0.758266$ holds throughout.

The value of $SU(6)$ by the previous lemma.

We then obtain $\sigma_c = 10.083(8)$ as claimed.

Recall that t_c in what follows.

It follows that $\tau_b \neq \tau_d$ for the remainder of the section.

Suppose $\phi(t, x)$ and the result follows.

Note that $\partial_t \phi(t, x)$ is used in the estimate below.

Here $\rho \propto (\partial_t \phi(t, x))^2 + ...$ holds throughout.

In this case $\vec{x} = \vec{x}_P$ by the previous lemma.

The value of $\partial_t \psi(t, \vec{x}_P)$ as claimed.

We then obtain $\psi(x, t_c)$ in what follows.

Recall that $\psi(t_0)$ for the remainder of the section.

It follows that $\psi(t) = e^{i\hat{H}(t-t_c)}\psi(t_c)$ and the result follows.

Suppose $\dot{q} = \infty$ is used in the estimate below.

Note that t_i holds throughout.

Here $dt < 0$ by the previous lemma.

In this case (t_a, t_f) as claimed.

The value of τ_d in what follows.

We then obtain (τ_c, τ_d) for the remainder of the section.

Recall that $q(\tau)$ and the result follows.

It follows that K_Z is used in the estimate below.

Suppose $dt = -d\tau$ holds throughout.

Note that $d\tau > 0$ by the previous lemma.

Here $\tau = \tau_c$ as claimed.

In this case $\tau_d - \tau_c$ in what follows.

The value of $\tau_c + n\epsilon$ for the remainder of the section.

We then obtain K_0 and the result follows.

Recall that $(\tau - \epsilon, \tau + \epsilon)$ is used in the estimate below.

It follows that $\tau_c + \epsilon, \tau_d$ holds throughout.

Suppose $\tau_c + N\epsilon = \tau_d$ by the previous lemma.

Note that t_{in} as claimed.

Here $\phi_p(t)$ in what follows.

In this case $T = \Delta T = \tau_c - \tau_b = \tau_d - \tau_c > 0$ for the remainder of the section.

The value of $n = 10$ and the result follows.

We then obtain $q_{\mathrm{F}} = 1 - \alpha$ is used in the estimate below.

Recall that 75% holds throughout.

It follows that $\mathrm{P}(\mathrm{ent}|Q) \geq q_{\mathrm{B}}$ by the previous lemma.

Suppose $\mathrm{P}(Q \in \mathcal{O}_{\mathrm{acc}}|\mathrm{sep}) \approx 0$ as claimed.

Note that σ_0 in what follows.

Here $R_{\mu_1} \cdots R_{\mu_N} = \pm 1$ for the remainder of the section.

In this case $\sigma_j^2 = 1 - T_j^2$ and the result follows.

The value of $n_j = n$ is used in the estimate below.

We then obtain 2.5% holds throughout.

Recall that 4.2% by the previous lemma.

It follows that $q \geq q_{\min}$ as claimed.

Suppose $E_{\mathrm{W}}^{\infty} = T_1 - T_2 - \cdots - T_M + 1$ in what follows.

Note that $n = 4$ for the remainder of the section.

Here $\mathrm{P}(\mathrm{ent} | E_{\mathrm{W}}) \geq 97.5\%$ and the result follows.

In this case $\mathrm{P}(S < \mathcal{B} | \mathrm{ent})$ is used in the estimate below.

The value of $N = 4$ holds throughout.

We then obtain $\mathrm{P}(\mathrm{sep}) = 1/17$ by the previous lemma.

Recall that $\bar{p} = 0.8$ as claimed.

It follows that $n_2 = 4$ in what follows.

Suppose 0.137 for the remainder of the section.

Note that $q_{\mathrm{B}} \geq 70\%$ and the result follows.

Here $\mathrm{G}_{\mathrm{acc}}(Q)$ is used in the estimate below.

In this case $\mathcal{O}_{\mathrm{acc}} = \{2.25, 3\}$ holds throughout.

The value of $E_{\mathrm{W}} < 0$ by the previous lemma.

We then obtain $(1 \pm T_i)/2$ as claimed.

Recall that $S = 1$ in what follows.

It follows that ρ_j for the remainder of the section.

Suppose $\bar{\rho}_{\mathrm{sep}} = \sum_j \rho_j \mathrm{P}(\rho_j | \mathrm{sep})$ and the result follows.

Note that $S^{\infty} \leq 1$ is used in the estimate below.

Here $10^{-8} \cdot \Phi_0$ holds throughout.

In this case T_2 by the previous lemma.

The value of $\{0, 1\}$ as claimed.

We then obtain τ_i in what follows.

Recall that $n = 0$ for the remainder of the section.

It follows that $\Phi / \Phi_0 \in [0, 0.5]$ and the result follows.

Suppose $\Delta = |\omega_q - \omega_r|$ is used in the estimate below.

Note that $\sim \omega_q^{-1/2}$ holds throughout.

Here M' by the previous lemma.

In this case $S_{\Phi}(\omega) = 2\pi A_{\Phi}^2 / \omega$ as claimed.

The value of $\delta \omega_q \simeq |\partial \omega_q / \partial I_c| \delta I_c$ in what follows.

We then obtain $\cdot 10^{-3}$ for the remainder of the section.

Recall that $\delta P_m \leq 1/(2\sqrt{K})$ and the result follows.

It follows that Φ^* is used in the estimate below.

Suppose $0 - 1$ holds throughout.

Note that $\Phi = \Phi_0/2$ by the previous lemma.

Here f_q as claimed.

In this case k_B in what follows.

The value of $\Phi^*/\Phi_0 = 0.442$ for the remainder of the section.

We then obtain $|ij\rangle$ and the result follows.

Recall that $\Gamma_N = N\Gamma$ is used in the estimate below.

It follows that $P_{|10\ldots0\rangle}$ holds throughout.

Suppose $M = 24$ by the previous lemma.

Note that $\overline{\tau_l}$ as claimed.

Here $\sigma_0 = \sigma_1 = 1.0$ in what follows.

In this case n_i for the remainder of the section.

The value of $\hat{\Phi}_{jkl}$ and the result follows.

We then obtain $10^{-8}\Phi_0$ is used in the estimate below.

Recall that $d\vec{l}$ holds throughout.

It follows that ω_m by the previous lemma.

Suppose $F_{\mathrm{FB}} = -gM\gamma\dot{x}$ as claimed.

Note that $1/\gamma$ in what follows.

Here n_c for the remainder of the section.

In this case $y_c(t)$ and the result follows.

The value of f_{fb} is used in the estimate below.

We then obtain $A[\omega]$ holds throughout.

Recall that $\omega_m/\gamma \gg 1$ by the previous lemma.

It follows that n_m as claimed.

Suppose $n_{\mathrm{add}} < 4$ in what follows.

Note that $A_0/2\pi = 6.7$ for the remainder of the section.

Here ω_f and the result follows.

In this case $\omega_m/2\pi = 8.14$ is used in the estimate below.

The value of $n_m^T \simeq 205$ holds throughout.

We then obtain $\Delta_f/2\pi = -20$ by the previous lemma.

Recall that $\delta f_2/2\pi \simeq 400$ as claimed.

It follows that $G/2\pi \simeq 104$ in what follows.

Suppose $n_m \simeq 2.9 \pm 0.3$ for the remainder of the section.

Note that X_{bax} and the result follows.

Here $x_c = \chi_c \sqrt{\kappa} x_{\mathrm{c,in}}$ is used in the estimate below.

In this case $a = \alpha(t) + \alpha_f(t) + \tilde{a}(t)$ holds throughout.

The value of $-\sqrt{2} g_0 a^\dagger a$ by the previous lemma.

We then obtain $\Delta_f + \omega_m$ as claimed.

Recall that $+\omega_m$ in what follows.

It follows that $s(t)$ for the remainder of the section.

Suppose $\cos s(t) \simeq 1$ and the result follows.

Note that ϕ_+ is used in the estimate below.

Here $D \equiv \sqrt{A^2 + B^2 + 2AB\cos(\varphi - \varphi')}$ holds throughout.

In this case $\omega_c + \omega_m + \Delta_f$ by the previous lemma.

The value of $arg(\chi_c[-\omega_f]) = -\phi_1$ as claimed.

We then obtain $\frac{1}{\sqrt{N}}$ in what follows.

Recall that $\frac{\hbar^2}{mL^2}$ for the remainder of the section.

It follows that $\hat{T}(x)$ and the result follows.

Suppose $\hat{W}(x_i - x_j)$ is used in the estimate below.

Note that $n_1 + n_2 + \cdots + n_M = N$ holds throughout.

Here $|n_1, n_2, \ldots, n_M\rangle$ by the previous lemma.

In this case $\{\phi_i(x,t)\}_{i=1}^{M}$ as claimed.

The value of $|\psi(t)\rangle$ in what follows.

We then obtain $C_{\bar{n}}(t)$ for the remainder of the section.

Recall that $E^N_{\lambda\to\infty} = \frac{N^2}{2}$ and the result follows.

It follows that 8.0 is used in the estimate below.

Suppose $x_{\max} = +32$ holds throughout.

Note that $\rho^{(1)}(x_1'|x_1;t)$ by the previous lemma.

Here $p = 2$ as claimed.

In this case $\rho^{(3)}(x_1, x_2, x_3 = x_{\mathrm{ref}}; t)$ in what follows.

The value of $|g^{(p)}(x_1, \ldots, x_p; t)| = 1$ for the remainder of the section.

We then obtain $x_1, \ldots, x_p$ and the result follows.

Recall that $g^{(2)}(0,0)$ is used in the estimate below.

It follows that $M = 1$ holds throughout.

Suppose $\sim 10^{-6}$ by the previous lemma.

Note that $R_{ic}(t)$ as claimed.

Here $R_c(t)$ in what follows.

In this case V_{ex}^{oc} for the remainder of the section.

The value of $V_{ex} = \frac{dR(t)}{dt}$ and the result follows.

We then obtain $N = 5$ is used in the estimate below.

Recall that $\rho^{(2)}(x_1, x_2)$ holds throughout.

It follows that $t = 0.0$ by the previous lemma.

Suppose 5.0 as claimed.

Note that $\rho^{(3)}(x_1, x_2, x_3 = 0.0)$ in what follows.

Here $x_1 = -x_2$ for the remainder of the section.

In this case $\mathbb{Z}_2$ and the result follows.

The value of $W^{\pm}, Z$ is used in the estimate below.

We then obtain $O(3)$ holds throughout.

Recall that $n = c^{\dagger}c$ by the previous lemma.

It follows that s_{xy}^{k} as claimed.

Suppose $V = 2t$ in what follows.

Note that $[Q_x, H^{\mathbb{Z}_2}] = 0$ for the remainder of the section.

Here $H^{U(1)}$ and the result follows.

In this case $H = H_e + H_o$ is used in the estimate below.

The value of $1, \cdots, 2N_t$ holds throughout.

We then obtain s_{xy}^{-} by the previous lemma.

Recall that w_A as claimed.

It follows that $[C_i]$ in what follows.

Suppose $(2+1)-$ for the remainder of the section.

Note that $G_x = 0$ and the result follows.

Here $\bar{\psi}\psi = \sum_x (-1)^x n_x$ is used in the estimate below.

In this case χ_{EE} holds throughout.

The value of 2×2 by the previous lemma.

We then obtain $m = 2^{N_f}$ as claimed.

Recall that $A^{\otimes N_f}$ in what follows.

It follows that $n = 2^{3N_f} - 2^{2N_f}$ for the remainder of the section.

Suppose $T \cdot N_t$ and the result follows.

Note that s_{i} is used in the estimate below.

Here n_{w} holds throughout.

In this case 10×10 by the previous lemma.

The value of 0.1 as claimed.

We then obtain $L = 8$ in what follows.

Recall that $\beta t = 0.1$ for the remainder of the section.

It follows that $L^{1.94}$ and the result follows.

Suppose $t_{2} = 0$ is used in the estimate below.

Note that c_{k} holds throughout.

Here ϵ_{k} by the previous lemma.

In this case $\mathcal{H}_{N_X}$ as claimed.

The value of $\mathbf{k}_{3}$ in what follows.

We then obtain $\mathbf{k}_{s}$ for the remainder of the section.

Recall that $R_{3}(t_{1}, t_{2}, t_{3})$ and the result follows.

It follows that $\mathcal{H}_{N_X \pm 1}$ is used in the estimate below.

Suppose $\mathcal{E}_{n}^{(P)}$ holds throughout.

Note that $|\eta\rangle_{B}$ by the previous lemma.

Here $\Phi_{0}^{(\eta)} \equiv \langle \mathrm{FS} | X_{0} X_{0} | \eta \rangle / \sqrt{2}$ as claimed.

In this case X_{0} in what follows.

The value of $\mathcal{S}_{\mathrm{ESA}}$ for the remainder of the section.

We then obtain $\phi_{0}^{(n=0)} = 1$ and the result follows.

Recall that $\mathcal{E}_{n}$ is used in the estimate below.

It follows that $n = N / La$ holds throughout.

Suppose $a = 1$ by the previous lemma.

Note that t_{c} / L as claimed.

Here $\mathcal{E}_{n} \to \mathcal{E}_{n} - nU$ in what follows.

In this case $t_{d} = 4\epsilon_{c}$ for the remainder of the section.

The value of $1/u$ and the result follows.

We then obtain $|\Phi_{0}^{(\eta)}|^{2}$ is used in the estimate below.

Recall that $\mathcal{E}_{B}$ holds throughout.

It follows that $-\omega_{1}$ by the previous lemma.

Suppose $(-\omega_{1}, \omega_{3}) \approx (\mathcal{E}_{A}, \mathcal{E}_{R})$ as claimed.

Note that $(-\omega_1, \omega_3)$ in what follows.

Here $k - k_p + k_h \geq -k$ for the remainder of the section.

In this case $|k_p k_h\rangle_1$ and the result follows.

The value of $z = 10$ is used in the estimate below.

We then obtain V_L holds throughout.

Recall that $a^2 \dot{D}_+$ by the previous lemma.

It follows that t_m as claimed.

Suppose M_2 in what follows.

Note that $v^2 \gg G(M_1 + M_2)/b$ for the remainder of the section.

Here $\langle L_i \rangle = 0$ and the result follows.

In this case $\mu = \hat{\boldsymbol{u}}.\hat{\boldsymbol{v}}$ is used in the estimate below.

The value of $|\mu|$ holds throughout.

We then obtain $\xi = P(|\mu|) - 1$ by the previous lemma.

Recall that $\hat{L}_i(n)$ as claimed.

It follows that $\mathrm{Tr}[A] = 1$ in what follows.

Suppose θ_a for the remainder of the section.

Note that $\cos \alpha_a = \hat{\boldsymbol{v}}.\hat{\boldsymbol{J^a}}$ and the result follows.

Here λ_i is used in the estimate below.

In this case $f(M, \boldsymbol{r}, \boldsymbol{v})$ holds throughout.

The value of $\boldsymbol{v}_0$ by the previous lemma.

We then obtain $M + dM$ as claimed.

Recall that $[\delta_{ij} + 3\hat{v_0}_i \hat{v_0}_j]$ in what follows.

It follows that m_{DM} for the remainder of the section.

Suppose 2160^3 and the result follows.

Note that Ω_Λ is used in the estimate below.

Here $\mu = \hat{\boldsymbol{L}}.\hat{\boldsymbol{v}}$ holds throughout.

In this case $a \gtrsim 1$ by the previous lemma.

The value of 0.04 as claimed.

We then obtain $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3$ in what follows.

Recall that n^3 for the remainder of the section.

It follows that $D_i = \sinh 2\pi i k_i / n$ and the result follows.

Suppose $|\cos \alpha| \sim 1$ is used in the estimate below.

Note that $\hat{\boldsymbol{J}}.\hat{\boldsymbol{e}}_2$ holds throughout.

Here $\mu_\nu = 0$ by the previous lemma.

In this case $\nu_R = (\nu_{eR}, \nu_{\mu R}, \nu_{\tau R})$ as claimed.

The value of H_L in what follows.

We then obtain $V_e = \sqrt{2} G_F (n_e - 0.5 n_n)$ for the remainder of the section.

Recall that ν_R and the result follows.

It follows that $\nu_{\alpha L} \leftrightarrow \nu_{\alpha R}$ is used in the estimate below.

Suppose $V_e + \dot{\phi} (> 0)$ holds throughout.

Note that r_0 by the previous lemma.

Here $B(r) = B_0 (r_0 / r)^3$ as claimed.

In this case Y_e in what follows.

The value of $\ddot{\phi}(r)$ for the remainder of the section.

We then obtain $r \lesssim 100$ and the result follows.

Recall that $r = r_0$ is used in the estimate below.

It follows that $x = \mu, \tau$ holds throughout.

Suppose $\mathrm{m}^{-1} \gtrsim \dot{\phi} \gtrsim -2$ by the previous lemma.

Note that $B_0 = 10^{12}$ as claimed.

Here ν_τ in what follows.

In this case 3×10^4 for the remainder of the section.

The value of $|U_{e3}| \approx 0.02$ and the result follows.

We then obtain $|U_{e1}|^2 \Phi_{ix}$ is used in the estimate below.

Recall that $\dot{\phi} > 0$ holds throughout.

It follows that 8.2×10^{-15} by the previous lemma.

Suppose $\nu_{\alpha L} \leftrightarrow \nu'_R$ as claimed.

Note that $15 M_\odot$ in what follows.

Here $\Phi_{ie} \to 0$ for the remainder of the section.

In this case Φ_e^{IO} and the result follows.

The value of 30% is used in the estimate below.

We then obtain 100% holds throughout.

Recall that N_t by the previous lemma.

It follows that F_i as claimed.

Suppose $f(R)$ in what follows.

Note that $\tilde{g}_{\mu\nu}$ for the remainder of the section.

Here $\{\psi\}$ and the result follows.

In this case $(-,+,+,+)$ is used in the estimate below.

The value of $-3\mu^4/2\lambda$ holds throughout.

We then obtain $\chi \ll M$ by the previous lemma.

Recall that $\tilde{\chi}^2/M^2$ as claimed.

It follows that $(\alpha_{hs}\tilde{\chi} + \lambda_{hs}\tilde{\chi}^2)\tilde{\phi}^2$ in what follows.

Suppose $\rho_\psi = y\tilde{\phi}_m\langle\bar{\tilde{\psi}}\tilde{\psi}\rangle$ for the remainder of the section.

Note that $\tilde{\phi}_m$ and the result follows.

Here $v^2 = 6\mu^2/\lambda$ is used in the estimate below.

In this case $1/M^2$ holds throughout.

The value of $\mu_\chi \ll \mu$ by the previous lemma.

We then obtain $\rho_\psi/(\mu^2 v^2)$ as claimed.

Recall that $\mu^2 v^2$ in what follows.

It follows that $A^2(\chi)$ for the remainder of the section.

Suppose $b = 1$ and the result follows.

Note that μ_χ is used in the estimate below.

Here $\tilde{\chi} = \tilde{\chi}_m + \delta\tilde{\chi}$ holds throughout.

In this case $\delta\tilde{\phi}$ by the previous lemma.

The value of $\mu_{\tilde{\chi}}^2/\mu^2$ as claimed.

We then obtain v^2/M^2 in what follows.

Recall that μ^2 for the remainder of the section.

It follows that $\sim \sqrt{\rho_\psi}/M$ and the result follows.

Suppose $\beta \sim 1$ is used in the estimate below.

Note that $\sim 10^{-26}$ g/cm^3 holds throughout.

Here ρ_ψ by the previous lemma.

In this case $\delta\chi = s\cos\theta$ as claimed.

The value of $r > R$ in what follows.

We then obtain x' for the remainder of the section.

Recall that x_{right} and the result follows.

It follows that $[n] = \{1, \ldots, n\}$ is used in the estimate below.

Suppose x_j holds throughout.

Note that $(c_1, \ldots, c_k) \in \mathbb{R}^k$ by the previous lemma.

Here $O(kn \log n)$ as claimed.

In this case $C_i = Y_i$ in what follows.

The value of $x_{i,r-1}$ for the remainder of the section.

We then obtain $\vec{C}'$ and the result follows.

Recall that $M(\vec{x}) = (y_1, ..., y_k)$ is used in the estimate below.

It follows that $\min_{j \in \{1,2,...,k\}}\{d(x, y_j)\} = d(x, y_i)$ holds throughout.

Suppose $\mathbb{P}\mathrm{r}[E]$ by the previous lemma.

Note that y_j as claimed.

Here $O(1)$ in what follows.

In this case $C'_{i'-1}$ for the remainder of the section.

The value of $d_{i,j}$ and the result follows.

We then obtain $x_{i-1,\hat{1}} \geq x_{i-1,\hat{2}} \geq \ldots \geq x_{i-1,\hat{n'}}$ is used in the estimate below.

Recall that X'_{i-1} holds throughout.

It follows that $> 0.4 \cdot D(C_i)$ by the previous lemma.

Suppose $d(x_{i,l}, c_{i-1}) > 5d(x_{i,l}, c_i)$ as claimed.

Note that $D(C_i)$ in what follows.

Here $\gamma \geq 1$ for the remainder of the section.

In this case (γ/γ') and the result follows.

The value of $d(x_i, x') > d(x_i, c_i)$ is used in the estimate below.

We then obtain $1 \leq i < j \leq n$ holds throughout.

Recall that $\{c_i, ..., x_{i,r}, ..., c_j\}$ by the previous lemma.

It follows that $d(x_{i,r}, x_{i+1,l}) > (\gamma - 1)d(x_{i,r}, c_i)$ as claimed.

Suppose $k - 2$ in what follows.

Note that $d(x_{i,l}, c_i) = y$ for the remainder of the section.

Here $[0, 2 - \sqrt{3}]$ and the result follows.

In this case $\mathbf{x}_i$ is used in the estimate below.

The value of $u(p)$ holds throughout.

We then obtain $\Lambda = \mathrm{diag}([\sigma_1^2, ...\sigma_d^2])$ by the previous lemma.

Recall that $\mathbb{E}(I^k)$ as claimed.

It follows that $K = 3$ in what follows.

Suppose $Q \in \mathbb{R}^{N \times K}$ for the remainder of the section.

Note that $\bar{P} = \frac{1}{M} \sum_{m=1}^{M} P_{m,*,*} \in \mathbb{R}^{N \times K}$ and the result follows.

Here $j = 1, ..., J$ is used in the estimate below.

In this case $2(N_l - 1)\Delta^2_{\mathcal{D}_l} \in \mathcal{F}$ holds throughout.

The value of 95.80 ± 1 by the previous lemma.

We then obtain 90.95 ± 2 as claimed.

Recall that 50.80 ± 2 in what follows.

It follows that 88.09 ± 2 for the remainder of the section.

Suppose 92.89 ± 2 and the result follows.

Note that 0.8 is used in the estimate below.

Here 87.06 ± 1.9 holds throughout.

In this case 91.44 ± 1.2 by the previous lemma.

The value of $\gamma \in \{0.5, 0.7, 0.9\}$ as claimed.

We then obtain $\{\mathbf{x}_i, y_i\}_{i=1}^{N_l}$ in what follows.

Recall that μ_- for the remainder of the section.

It follows that $\tilde{X}_i^- \sim \mathcal{N}(\mu_-, \sigma^2)$ and the result follows.

Suppose $\texttt{Var}(I_i^+)$ is used in the estimate below.

Note that $2(N_l - 1)\Delta_{\mathcal{D}}^2 \in \mathcal{F}$ holds throughout.

Here $1, ...K$ by the previous lemma.

In this case 95.74 ± 1 as claimed.

The value of 57.50 ± 2 in what follows.

We then obtain 81.03 ± 3 for the remainder of the section.

Recall that $T \leq 5$ and the result follows.

It follows that $\mathit{89.14} \pm 2$ is used in the estimate below.

Suppose $\mathit{82.91} \pm 3$ holds throughout.

Note that h_z by the previous lemma.

Here N_p as claimed.

In this case $p_k \in \{+1, -1\}$ in what follows.

The value of $\mathcal{E}' = \{e'_k\}_{k=1}^{N_e}$ for the remainder of the section.

We then obtain $b_k = p_k$ and the result follows.

Recall that $G(\boldsymbol{\theta})$ is used in the estimate below.

It follows that s_k holds throughout.

Suppose $h_z < 0$ by the previous lemma.

Note that $\nabla \cdot \mathbf{v} = \partial_x \mathbf{v}_x + \partial_y \mathbf{v}_y$ as claimed.

Here $N_e(\mathbf{x}) \doteq \sum_k \delta(\mathbf{x} - \mathbf{x}'_k)$ in what follows.

In this case $|\det(\mathtt{J})| < 1$ for the remainder of the section.

The value of $|\det(\mathtt{J})| \in [0, \infty)$ and the result follows.

We then obtain $\det(\mathtt{J}(e_k)) = s_k^2$ is used in the estimate below.

Recall that $\mathbf{x}_p$ holds throughout.

It follows that $\mathbf{x}, t$ by the previous lemma.

Suppose $\boldsymbol{\theta} \equiv \boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)^\top$ as claimed.

Note that $\det(\mathsf{J}) = 1$ in what follows.

Here $R(\boldsymbol{\theta})$ for the remainder of the section.

In this case $dt = 4$ and the result follows.

The value of $\lambda_{\text{def}} = 5$ is used in the estimate below.

We then obtain > 80 holds throughout.

Recall that 10–20 by the previous lemma.

It follows that $h_z \approx 0$ as claimed.

Suppose $\mathbf{x}(t)$ in what follows.

Note that $\boldsymbol{\theta} = \boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)^\top$ for the remainder of the section.

Here $t_k \|\boldsymbol{\omega}\|$ and the result follows.

In this case $\det(\mathsf{J}_k) \approx 1 + 3(x_k \omega_2 - y_k \omega_1) t_k$ is used in the estimate below.

The value of $\mathbf{x}_k' = (1 - t_k h_z)\mathbf{x}_k$ holds throughout.

We then obtain $SO(2)$ by the previous lemma.

Recall that $\beta_k = 1 + t_k s$ as claimed.

It follows that T_K in what follows.

Suppose w_0 for the remainder of the section.

Note that $T = 20$ and the result follows.

Here 0.432 ± 0.018 is used in the estimate below.

In this case 1.918 ± 0.006 holds throughout.

The value of 1.912 ± 0.021 by the previous lemma.

We then obtain 0.485 ± 0.020 as claimed.

Recall that 1.927 ± 0.006 in what follows.

It follows that 5.596 ± 0.030 for the remainder of the section.

Suppose 4.597 ± 0.070 and the result follows.

Note that $\mathcal{P}, \mathcal{Q}$ is used in the estimate below.

Here $\mathcal{F}^{-1}$ holds throughout.

In this case $K^d \times \mathrm{C} \times \mathrm{C}$ by the previous lemma.

The value of $K = 90$ as claimed.

We then obtain s_t in what follows.

Recall that $K_t \leftarrow K_t + 1$ for the remainder of the section.

It follows that (K_t, R_{curr}) and the result follows.

Suppose $g(K, s_p) > \alpha$ is used in the estimate below.

Note that K_{t+1} holds throughout.

Here 918 ± 10 by the previous lemma.

In this case 0.155 ± 0.003 as claimed.

The value of 13.53 ± 13.08 in what follows.

We then obtain 2.844 ± 0.070 for the remainder of the section.

Recall that 0.276 ± 0.083 and the result follows.

It follows that 0.316 ± 0.015 is used in the estimate below.

Suppose 1.130 ± 0.023 holds throughout.

Note that 1.290 ± 0.022 by the previous lemma.

Here 5.627 ± 0.073 as claimed.

In this case x_1 in what follows.

The value of 128×128 for the remainder of the section.

We then obtain $f : X \to \mathbb{R}$ and the result follows.

Recall that $\mathsf{Hom}(-, \mathbb{R}_+)$ is used in the estimate below.

It follows that (F, μ, η) holds throughout.

Suppose $a \in X$ by the previous lemma.

Note that $[\phi] = 1$ as claimed.

Here (D, μ, η) in what follows.

In this case $\mu^D : D^2 \to D$ for the remainder of the section.

The value of $\mathbb{R}_+$ and the result follows.

We then obtain $\otimes^L : LF \to FL$ is used in the estimate below.

Recall that $[\phi]$ holds throughout.

It follows that $\otimes_X^L$ by the previous lemma.

Suppose $\otimes^L : LD \to DL$ as claimed.

Note that $(\cdot)$ in what follows.

Here $f : X \to Y$ for the remainder of the section.

In this case $\{1, \ldots, n\}$ and the result follows.

The value of $\otimes_X^M$ is used in the estimate below.

We then obtain $d_1, \ldots, d_n$ holds throughout.

Recall that $\phi' : F' \to F''$ by the previous lemma.

It follows that $\phi : F \to G$ as claimed.

Suppose $\phi : \Phi \to \Psi$ in what follows.

Note that ϕ_S for the remainder of the section.

Here $I_O : O \to O$ and the result follows.

In this case $\tau : T \to U$ is used in the estimate below.

The value of $\sigma, \tau : S \to T$ holds throughout.

We then obtain $\Psi\sigma = \Psi\tau$ by the previous lemma.

Recall that $\mu^M : MM \to M$ as claimed.

It follows that $\otimes_X^L(\bar{f})(\bar{a}) = 0$ in what follows.

Suppose $\mathsf{supp} \otimes_X^L(\bar{f}) = \prod_{i=1}^{n} \mathsf{supp}\, f_i$ for the remainder of the section.

Note that $Fh(f) \in FY$ and the result follows.

Here $f_i \in FFX = \mathbb{R}^{\mathbb{R}^X}$ is used in the estimate below.

In this case $p(\mathbf{X})$ holds throughout.

The value of $\{X_i, Z_i, Y_i, R_i\}_{i=1}^{N_s}$ by the previous lemma.

We then obtain R_i as claimed.

Recall that $N_{pop} >> N_s$ in what follows.

It follows that $D(\cdot, \cdot)$ for the remainder of the section.

Suppose $\mu = \frac{1}{N_{pop}} \sum_i Y_i$ and the result follows.

Note that $g(\cdot) : \mathcal{R} \mapsto [0, 1]$ is used in the estimate below.

Here $\phi(X)^\top \beta + (Z + \nu)$ holds throughout.

In this case N_s by the previous lemma.

The value of $k(X_i, X_j)$ as claimed.

We then obtain $\phi(X_i)^\top \theta = \sum_j c_j k(X_i, X_j)$ in what follows.

Recall that i^{th} for the remainder of the section.

It follows that $X_i = X_j$ and the result follows.

Suppose l^{th} is used in the estimate below.

Note that $\mathbf{K} = \mathbf{VAU}^\top$ holds throughout.

Here $(w^{(pop)\top} \mathbf{V}_{pop} - w_s^\top \mathbf{V}_s)$ by the previous lemma.

In this case $exp(-||X_i - X_j||^2/b)$ as claimed.

The value of $k(X_i, X_j) = e^{-\left((1-0)^2 + (0-0)^2\right)} = e^{-1} \approx 0.37$ in what follows.

We then obtain $k(, 2)$ for the remainder of the section.

Recall that $N_s = 2,052$ and the result follows.

It follows that $kpop + mf$ is used in the estimate below.

Suppose $\{X_i, Z_i, Y_i, R_i\}$ holds throughout.

Note that $Y - \phi(X)^\top \beta$ by the previous lemma.

Here $\mu_s(w)$ as claimed.

In this case $p(X = x)$ in what follows.

The value of $\phi(x)$ for the remainder of the section.

We then obtain $g(\cdot) : \mathbb{R} \mapsto [0, 1]$ and the result follows.

Recall that $\phi(X) = X = [\text{female},\ \text{college}]$ is used in the estimate below.

It follows that $b = 2D$ holds throughout.

Suppose R^2 by the previous lemma.

Note that $X_{ij}, j = 1, \ldots, p$ as claimed.

Here c_j^2 in what follows.

In this case $\delta_0(.)$ for the remainder of the section.

The value of $\mathbf{B}_g$ and the result follows.

We then obtain $\omega_{gj,k}$ is used in the estimate below.

Recall that $IG(.)$ holds throughout.

It follows that s^2 by the previous lemma.

Suppose $Y_{i,k}$ as claimed.

Note that $\Phi^{-1}(.)$ in what follows.

Here τ^2 for the remainder of the section.

In this case $\text{sgn}(\cdot)$ and the result follows.

The value of $n \to \infty$ is used in the estimate below.

We then obtain $d_{gj}(t) = I\{P_{gj}(H_0|\mathbf{X}, \mathbf{Y}) < t\}$ holds throughout.

Recall that $PIP_{gj}^{\mathcal{M}_\mathcal{S}}$ by the previous lemma.

It follows that $\mathcal{M}_\mathcal{S}$ as claimed.

Suppose $r \leftarrow 1$ in what follows.

Note that σ_Y^2 for the remainder of the section.

Here $n = 500$ and the result follows.

In this case $\beta_{gj,k} = 0$ is used in the estimate below.

The value of n_{test} holds throughout.

We then obtain $\vec{M}_{gj}$ by the previous lemma.

Recall that $k \neq k'$ as claimed.

It follows that Σ_{kk}^{-1} in what follows.

Suppose $\xi_{gj}|\gamma_{gj}, d_0, d_1 \sim N(d_0 + d_1 A_{gj}, 1)_-$ for the remainder of the section.

Note that (σ^2, τ^2) and the result follows.

Here $\mathbf{X}^T\mathbf{X} = n\mathbf{I}$ is used in the estimate below.

In this case $M = (1 - D_n)\hat{\beta}^{LS}_{gj,k}$ holds throughout.

The value of $\hat{\beta}^{LS}_{gj,k} < 0$ by the previous lemma.

We then obtain $\sqrt{n}|\hat{\beta}^{LS}_{gj,k}| = O_p(1)$ as claimed.

Recall that $l_{gj,k} \to 1$ in what follows.

It follows that τ_b for the remainder of the section.

Suppose $\mathcal{Z} \subseteq \mathcal{V}$ and the result follows.

Note that $\mathbf{Y}_i$ is used in the estimate below.

Here $\mathcal{D} = (\mathcal{T}, h)$ holds throughout.

In this case $Z_5 = Z_6 = d$ by the previous lemma.

The value of $|\mathcal{Z}|$ as claimed.

We then obtain $u, v \in P_{m-1}$ in what follows.

Recall that $b = \min\{h(v) - h(\mathrm{Pa}_v)\}$ for the remainder of the section.

It follows that $\max_{v \in \mathcal{Z}} \|\mathbf{S}(v)\|_{\mathrm{op}} \in O(1)$ and the result follows.

Suppose $O_{\mathbb{P}}(\cdot)$ is used in the estimate below.

Note that $p^{-1}\mathbb{E}[\|\mathbf{Y}_i\|^2 | Z_1, \ldots, Z_n]$ holds throughout.

Here $n = 5079$ by the previous lemma.

In this case $O(npr)$ as claimed.

The value of $\mathbf{Y}_1, \ldots, \mathbf{Y}_n \in \mathbb{R}^p$ in what follows.

We then obtain $u \neq v$ for the remainder of the section.

Recall that $w' \in P_m$ and the result follows.

It follows that P_0 is used in the estimate below.

Suppose $\hat{h}(w_m) = \hat{\alpha}(u_m, v_m) \geq \hat{h}(w_{m+1})$ holds throughout.

Note that $l \in B$ by the previous lemma.

Here z_l as claimed.

In this case $S_v \neq [n]$ in what follows.

The value of $A' \subseteq S_v$ for the remainder of the section.

We then obtain $m^* + 1$ and the result follows.

Recall that $(\mathcal{Z}, d_{\mathcal{Z}})$ is used in the estimate below.

It follows that Δ_1 holds throughout.

Suppose $\mathbf{E}_j \in \mathbb{R}^p$ by the previous lemma.

Note that $|\mathcal{Z}| - 1$ as claimed.

Here $p = 1000$ in what follows.

In this case $d_{\cos}(\cdot, \cdot)$ for the remainder of the section.

The value of $\hat{\alpha}_{\cos}(i, j)$ and the result follows.